Fall 2026 Section: _____ Name: _____

Read the following directions carefully. DO NOT turn to the next page until the exam has started.

Write your name and section number at the top right of this page:

Class	Section Number
8:50	001
9:55	004

As you complete the exam, write your initials at the top right of each other page.

When the exam start time is called, you may turn the page and begin your exam.
If you need more room, there is a blank page at the end of the exam, or we can give you some scratch paper.

Some multiple choice questions are “Select ONE" while others are “Select ALL that apply". Pay attention to the question type and only mark one option if it says “Select ONE". Fill in the circles completely. Only highlight the circles of the selected options and leave the others completely blank.

If you finish early, you can hand your exam to your instructor or TA and leave early.

Otherwise, stop writing and hand your exam to your instructor or TA when the exam stop time is called.

1. The R dataset `mtcars` records fuel consumption and design aspects for 32 types of automobiles from 1974. The first six rows are shown below, and there are no missing values anywhere in the data.

```
> head(mtcars)
                   mpg cyl disp  hp drat    wt  qsec vs am gear carb
Mazda RX4         21.0   6  160 110 3.90 2.620 16.46  0  1    4    4
Mazda RX4 Wag     21.0   6  160 110 3.90 2.875 17.02  0  1    4    4
Datsun 710        22.8   4  108  93 3.85 2.320 18.61  1  1    4    1
Hornet 4 Drive    21.4   6  258 110 3.08 3.215 19.44  1  0    3    1
Hornet Sportabout 18.7   8  360 175 3.15 3.440 17.02  0  0    3    2
Valiant           18.1   6  225 105 2.76 3.460 20.22  1  0    3    1
```

   (a) (4 points) The column `am` has value 0 for an automatic transmission car and 1 for a manual transmission. Which lines of R code can be used to output the proportion of manual transmission cars in the data? **Select ALL that apply**.

   - [x] `mean(mtcars$am)`
   - [ ] `sum(mtcars$am)`
   - [ ] `mean(mtcars$am = 1)`
   - [x] `mean(mtcars$am == 1)`

   (b) (6 points) You would like to find the average engine displacement (`disp`) for cars with mpg between 20 and 25, based on the number of cylinders (`cyl`). Code to perform this task is given below, but it has several errors.

```
mtcars +
    filter(mpg >= 20 | mpg <= 25) %>%
    group_by(cyl) %>%
    summarize(n = n(), Total = sum(disp), Avg_Disp = avg(disp))
```

   What changes should be made to fix this code? **Select ALL that apply**.

   - [x] Change the + to %>% in the first line.
   - [ ] Change `group_by()` to `select()`.
   - [ ] Change `filter()` to `select()`.
   - [x] Change `avg(disp)` to `mean(disp)` inside `summarize()`.
   - [ ] Change `Avg_Disp` to `mean_Disp` inside `summarize()`.
   - [x] Change the `|` symbol inside `filter()` to `&`.

   (c) (3 points) Explain what the code `n = n(), Total = sum(disp)` does inside of the `summarize()` function. Is this a necessary step in order to calculate the average displacement per group?

> **Solution:** The code `n = n()` calculates the total number of rows per group and `Total = sum(disp)` calculates the total of all values in the `disp` column for each group. These steps can be used to calculate the group means by hand, but they are not necessary.

2. A friend is studying a dataset that records several variables on patients at a hospice facility. He wants to use `ggplot()` to create several different visualizations of the data.

   (a) (4 points) Match each scenario your friend wants to investigate to the most appropriate geom. Choose from the list provided (you may not need to use every one)

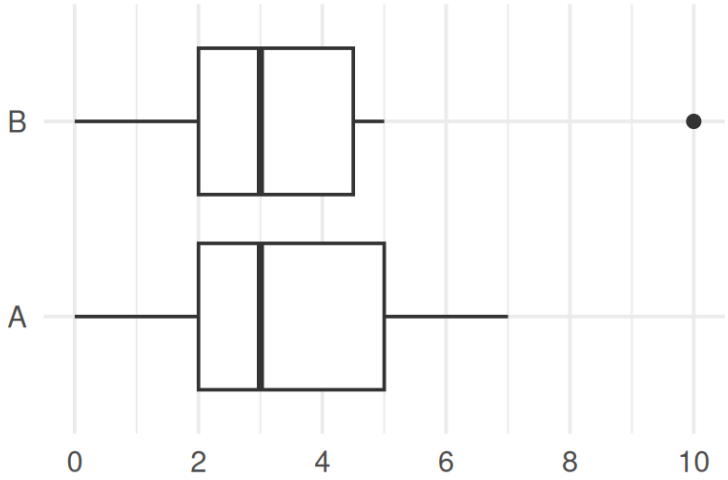
   {`geom_smooth`, `geom_bar`, `geom_col`, `geom_line`, `geom_point`, `geom_histogram`, `geom_boxplot`}

   - He suspects the distribution of patient ages is bimodal (concentrated around two different values). To investigate this, he should use _____`geom_histogram`_____.
   - He would like to categorize the patients by type of care received and draw bars showing the median length of stay for patients in each category. He should use _____`geom_col`_____.
   - He wants to investigate the relationship between patient age and blood pressure, but he only wants to see a trend without the noise from specific patients. He should use _____`geom_smooth`_____.
   - The friend summarizes the data to show the total number of patients at the facility per year. To make a connected time-series graph showing how the number of patients changes over time, he should use _____`geom_line`_____.

   (b) (3 points) Consider the fourth scenario described in part (a). Your friend wants to add points to the plot and have different-colored points for years when the COVID-19 pandemic was active. However, he wants there to still be a single connected line in the time series. How can this be done? **Select ONE**.

   ◯ `x`, `y`, and `color` should all be global variable aesthetics.
   ● `x` and `y` should be global variable aesthetics, and `color` should be a local variable aesthetic.
   ◯ `x` and `y` should be local variable aesthetics, and `color` should be a constant aesthetic.
   ◯ None of the above; this type of plot is not possible with `ggplot()`.

3. The plot below shows side-by-side boxplots of two groups, A and B.



(a) (3 points) How many *variable* aesthetics are required to create the plot above with 'ggplot()'? You can assume the graph does not have color. **Select ONE**.

- ○ 0
- ○ 1
- ● 2
- ○ 3

(b) (4 points) Fill in the blanks with "less than", "equal to" or "greater than" to complete the four statements below.

- The median of B is ___less than___ the mean of B.
- The median of A is ___equal to___ the median of B.
- the range of A is ___less than___ the range of B.
- The interquartile range of A is ___greater than___ the interquartile range of B.

(c) (4 points) The 25th percentile of group B is 2 and the 75th percentile of group B is 4.5. What are the numerical cutoffs for outliers (both lower and upper) in group B? You do not need to simplify your answer and can leave it as an expression.

Solution: The IQR, i.e. the length of the box, is $4.5 - 2 = 2.5$ For a point to be considered an outlier, it must be $1.5 \times IQR$ units away from the box, or 1.5(2.5). So, any point below $2 - 1.5(2.5)$ is considered a lower outlier and any point above $4.5 + 1.5(2.5)$ is considered an upper outlier.

(d) (3 points) Why are the "whiskers" on the side of the boxplot for B not drawn all the way out to the cutoffs you calculated?

Solution: The whiskers/lines on the side of the box plot will only be drawn out to the point where data actually exists. The left-most whisker is drawn to the minimum value of 0. The right-most one is drawn up to 5, which is the largest value that is not an outlier.

4. A dataframe `lions` has 32 rows, each representing a unique lion.

   It has three total columns. There are two numeric columns: `age`, each lion's age in decimal years, and `proportion.black`, the percentage of that lion's nose which is black. There is one logical column `adult`, which is `TRUE` if `age` is greater than 3 and `FALSE` otherwise.

   (a) (4 points) Consider the following code which creates a new dataframe, `lions_summary`.

```
lions_summary <- lions %>%
 group_by(adult) %>%
 summarize(averageAge = mean(age),
 averagePropBlack = mean(proportion.black))
```

   Which of the following statements are true about `lions_summary`? **Select ALL that apply**.

   - [x] `lions_summary` has 2 rows.
   - [ ] `lions_summary` has 2 columns.
   - [x] `lions_summary` has a column called "`adult`".
   - [ ] If there were any `NA` values in the `age` column of `lions`, all the values of `averageAge` in `lions_summary` will be `NA`.

   (b) (3 points) Now, consider the following code which is similar to the code in (a) but uses `mutate()` instead of `summarize()`.

```
lions_new <- lions %>%
 group_by(adult) %>%
 mutate(averageAge = mean(age),
 averagePropBlack = mean(proportion.black))
```

   How many rows will be in the resulting dataframe `lions_new`? How many *unique* values of `averageAge` will there be? Briefly explain your answers.

   Solution: There will still be 32 rows, just like the original data, since we are mutating a new column and not summarizing. There will be two unique values for `averageAge`, with each row being assigned a different value in that column according to whether the row belongs to the `adult == TRUE` group or the `adult == FALSE` group.

5. The 8×3 dataframe `students` records the ID, major, and graduating year for 8 undergraduate students. The 5×2 dataframe `aid` records the ID and type of aid received by 5 students. The 6×3 dataframe `school` records 6 departments, each one's founding year, and whether each one requires a minor.

```
> students
# A tibble: 8 × 3
  id    major       year
  <chr> <chr>       <chr>
1 OV480 Mathematics 2027
2 SL652 CS          2028
3 IB302 Statistics  2028
4 NY119 CS          2029
5 QA397 Undecided   2029
6 LK849 CS          2030
7 NN752 History     2027
8 DB296 Chemistry   2029
```

```
> aid
# A tibble: 5 × 2
  id    type
  <chr> <chr>
1 NN752 grant
2 LK849 scholarship
3 NY119 scholarship
4 GP233 work-study
5 OV480 grant
```

```
> school
# A tibble: 6 × 3
  department  year  req_minor
  <chr>       <chr> <lgl>
1 Mathematics 1905  TRUE
2 CS          1992  FALSE
3 Statistics  1974  FALSE
4 History     1905  TRUE
5 Chemistry   1905  FALSE
6 Biology     1919  FALSE
```

Consider performing the following join operation on the three datasets.

```
students %>%
    semi_join(aid, by = join_by(id)) %>%
    left_join(school, by = join_by(major == department))
```

(a) (4 points) Fill in the blanks: The resulting dataframe will have ____4____ rows and ____5____ columns.

(b) (4 points) Which columns, if any, will have missing/`NA` values? Briefly explain your answer.

Solution: Since we are doing a left join, all of the rows from the `students` data will be present. Since we are joining by major/department, the extra columns from `school` (`year` and `req_minor` would have missing data for the student with an Undecided major, since that does not have a corresponding department in the `school` data. However, since that row was filtered out by the `semi_join` in the previous step, all of the remaining students will have valid majors and present in the `school` data and so there will be no missing values.

(c) (3 points) Suppose the `left_join` in the above code is done with `join_by(year)` instead of `join_by(major == department)`. How, if at all, will this change the number of rows in the resulting dataframe? How, if at all, will this change the number of rows with `NA` values in the dataframe? Briefly explain your answers.

Solution: The number of rows will not change since it is still a left join and will keep exactly all of the rows from the original `students` data. However, the `major` and `department` columns will no longer be joined correctly, and none of the `years` between the data will match, so all of the rows will have missing data for `department` and `req_minor`.

```
new_data <- function(data, n) {
  sample_n(data, size = n, replace = TRUE)
}
```

6. Scientists record the lifespan of 85 captive fruit flies (in days) in the dataset `flies` under the `lifespan` column. Consider using this data to perform a resampling experiment to study the mean fruit fly lifespan.

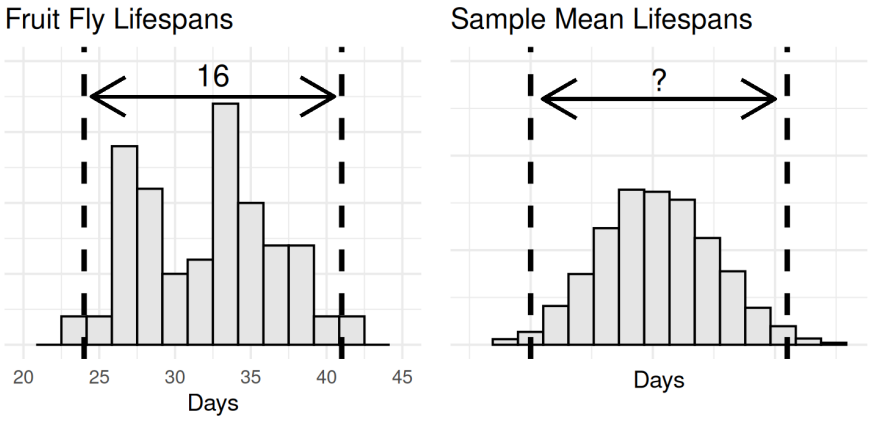
   (a) (4 points) Fill in the blanks to complete the code to resample from the data to construct a histogram of the sampling distribution of the mean of 85 fly lifespans. You should have 10000 resamples in your experiment. The function `new_data()` is given at the top of the page and is exactly the same as the one given to you in lecture and on homeworks.

```
tibble(
    xbars = replicate(  10000  ,
            mean  (new_data(flies,  85  )$lifespan))
)
```

   (b) (4 points) Select all true statements about this resampling process. **Select ALL that apply**.

   - ◯ There are 85 sample means, representing 85 theoretical fruit fruit fly experiments.
   - ● There are 10000 sample means, representing 10000 theoretical fruit fruit fly experiments.
   - ◯ Each time we resample, we are gathering a new set of data from the population of fruit flies.
   - ● Each time we resample, we might measure the same object twice.

   (c) (3 points) A histogram of the original 85 fruit fly lengths (left) and a histogram of the resampled means from the previous part (right) are given below. The x and y axes are NOT the same between the two histograms.



   The distance between the dashed lines on the original data histogram is about 16 days. Which is closest to the distance between the dashed lines on the histogram of resampled means? **Select ONE**.

   ● 2 days ◯ 16 days ◯ 128 days

   (d) (3 points) Consider building a confidence interval for the average fruit fly lifespan. The mean lifespan in the data is $M_{obs} = 32.04$. What percentiles of the resampled means should we use for M_{lower} and M_{upper} if we want to build a 99% confidence interval? **Select ONE**.

   - ◯ M_{lower} is the 99.5th percentile, M_{upper} is the 0.5th percentile
   - ◯ M_{lower} is the 99th percentile, M_{upper} is the 1st percentile
   - ● M_{lower} is the 0.5th percentile, M_{upper} is the 99.5th percentile
   - ◯ M_{lower} is the 1st percentile, M_{upper} is the 99th percentile

7. A teacher comes up with a puzzle to give to their students for extra credit, and their goal is for 60% of students to be able to answer it correctly.

   (a) (3 points) Let p be the true probability that a random student solves the puzzle correctly. Which hypotheses can the teacher use to answer their question? **Select ONE**.

   - [x] $H_0 : p = 0.6$ versus $H_A : p \neq 0.6$
   - [ ] $H_0 : p \neq 0.6$ versus $H_A : p = 0.6$
   - [ ] $H_0 : p = 0.6$ versus $H_A : p < 0.6$
   - [ ] $H_0 : p = 0.6$ versus $H_A : p > 0.6$

   (b) (3 points) A resampling experiment is performed, and 10000 sample proportions of students who correctly solved the puzzle are stored in the vector `resampled_props`.

   64% of the teacher's actual students solved the puzzle correctly. Which line of R code can be used to calculate the p-value for the test in (a)? **Select ONE**.

   - [ ] `mean(resampled_props <= 0.4) + mean(resampled_props >= 0.6)`
   - [x] `mean(resampled_props <= 0.56) + mean(resampled_props >= 0.64)`
   - [ ] `mean(resampled_props <= 0.36) + mean(resampled_props >= 0.64)`
   - [ ] `mean(resampled_props <= 0.6) + mean(resampled_props >= 0.68)`

   (c) (5 points) The p-value is about 0.45, so the teacher fails to reject their null hypothesis. Which of the following are valid conclusions from this test? **Select ALL that apply**.

   - [ ] The true probability of a student solving the puzzle correctly is 0.6.
   - [x] We lack evidence that the true probability is different from 0.6.
   - [ ] The teacher should choose an α above 0.45 so they can instead reject the null hypothesis.
   - [ ] We could have made a Type I error.
   - [x] We could have made a Type II error.